Take-Home Assignment week 37–38

Statik & Styrkelære

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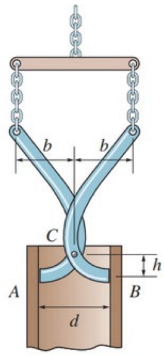
Exercise P5-1

The pipe is hoisted using the tongs. If the coefficient of static friction at A and B is μ_s, determine the smallest dimension b so that any pipe of inner diameter d can be lifted.



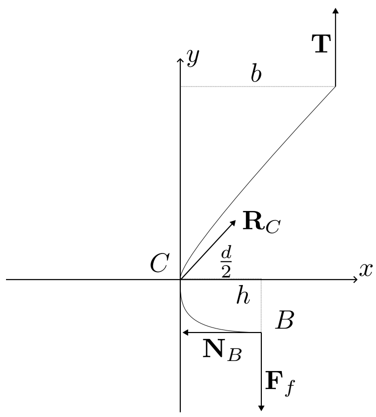


Figure 0.1: Free-body diagram of the right tong half

We consider one half of the tong, in this case the right half as shown on Figure 0.1. For this to be in equilibrium both the vector sum of all forces and moments about any point must be zero. I.e.

$$\mathbf{F}_R = \sum \mathbf{F} = \mathbf{0}$$
$$\mathbf{M}_O = \sum \mathbf{M}_O = \mathbf{0}.$$

For a couple moment not to arise, the lines of actions of all the forces must intersect at some common point, P, i.e. the forces must be concurrent. If the forces are not concurrent, there will always be some "perpendicular distance" between the lines of action of the forces and therefore a couple moment might be produced even if force equilibrium is required.

At point B, right at the verge of slippage, the line of action of the resultant force at B (which is the sum of the friction force and normal force components at this point) makes the friction angle, $\phi = \tan^{-1} \mu_s$ below the horizontal. We can therefore find the equation of the line using the formula for a line through a point, which in this case is $\left(-h, \frac{d}{2}\right)$ and a slope, which in this case is μ as:

$$y + h = \mu_s(x - \frac{d}{2}) \implies y = \mu_s\left(x - \frac{d}{2}\right) - h.$$

The line of action at point B will lie along this line.

At the top the tong is attached to a vertical chain, such that the force that is transmittet to the tong at the top is vertical, with a line of action of $x = b$.

As mentioned above, all the lines of actions of the forces must intersect at a common point P. This means that the two lines defined above must intersect at this point P. The intersection between the line of action of $\mathbf{T}$ and the line of action of $\mathbf{R}_B$ can now be found as:

$$P = \left(b, \mu_s \left(b - \frac{d}{2}\right) - h\right).$$

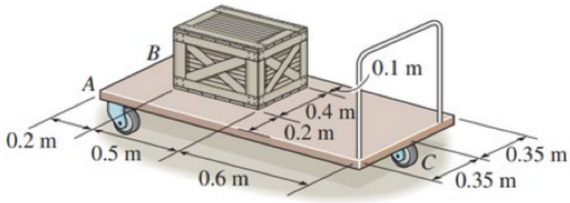
The pin force $\mathbf{R}_C$ at point C will act along the line CP. If this point P is below point C one can imagine that the tongues will "open up" and not press against the wall. Therefore, for the jaws to wedge and hold, the concurrency point P must be at or above C as:

$$\begin{aligned} y_p &\geq 0 \\ \mu_s \left(b - \frac{d}{2}\right) - h &\geq 0 \\ \implies \mu_s \left(b_{\min} - \frac{d}{2}\right) - h &= 0 \\ b_{\min} &= \frac{h}{\mu_s} + \frac{d}{2}. \end{aligned}$$

Therefore the smallest dimension of b such that the pipe can be lifted is $\underline{\underline{b_{\min} = \frac{h}{\mu_s} + \frac{d}{2}}}$.

Exercise P6-1

The cart supports the uniform crate having a mass of 85 kg. Determine the vertical reactions on the three casters at A, B, and C. The caster at B is not shown. Neglect the mass of the cart.



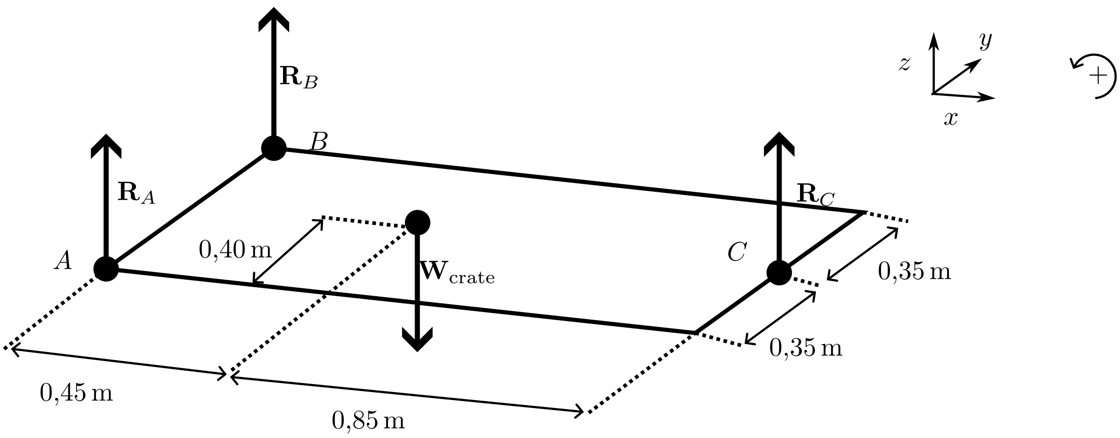


Figure 0.2: Free body diagram

A free body diagram of the cart us shown on Figure 0.2. For the cart to be in equilibrium the equilibrium equations must be satisfied. In three dimensions these may be:

$$\sum F = R_A + R_B + R_C + W_{\text{crate}} = 0$$
$$\sum (M_A)_x = |AB|_x \, R_B + |AW|_x \, W_{\text{crate}} + |AC|_x \, R_C$$
$$\sum (M_A)_y = |AB|_y \, R_B + |AW|_y \, W_{\text{crate}} + |AC|_y \, R_c.$$

First we find the size of $\mathbf{W}_{\text{crate}}$ using Newton's second law as

$$W_{\text{crate}} = 85\,\text{kg} \cdot \left(-9{,}81\,\frac{\text{m}}{\text{s}^2}\right) = -833{,}85\,\text{N}.$$

This weight will be acting from the center of the crate as it is assumed to be homogeneous. The distances can be read of the free body diagram to be

$$\begin{aligned} |AB|_x &= 0 \\ |AW|_x &= 0{,}45\,\mathrm{m} \\ |AC|_x &= 1{,}30\,\mathrm{m} \\ |AB|_y &= 0{,}70\,\mathrm{m} \\ |AW|_y &= 0{,}40\,\mathrm{m} \\ |AC|_y &= 0{,}35\,\mathrm{m}. \end{aligned}$$

Hence

$$\begin{aligned} R_A + R_B + R_C &= 833{,}85\,\mathrm{N} \\ 1{,}30\,\mathrm{m} \cdot R_C &= 0{,}45\,\mathrm{m} \cdot 833{,}85\,\mathrm{N} \\ 0{,}70\,\mathrm{m} \cdot R_B + 0{,}35\,\mathrm{m} \cdot R_C &= 0{,}40\,\mathrm{m} \cdot 833{,}85\,\mathrm{N}. \end{aligned}$$

Solving for R_C can easily be done as

$$R_C = \frac{0{,}45\,\mathrm{m} \cdot 833{,}85\,\mathrm{N}}{1{,}30\,\mathrm{m}} = 288{,}64\,\mathrm{N}.$$

Now, solving for R_B becomes trivial as

$$R_B = \frac{0{,}40\,\mathrm{m} \cdot 833{,}85\,\mathrm{N} - 0{,}35\,\mathrm{m} \cdot 288{,}64\,\mathrm{N}}{0{,}70\,\mathrm{m}} = 332{,}17\,\mathrm{N}.$$

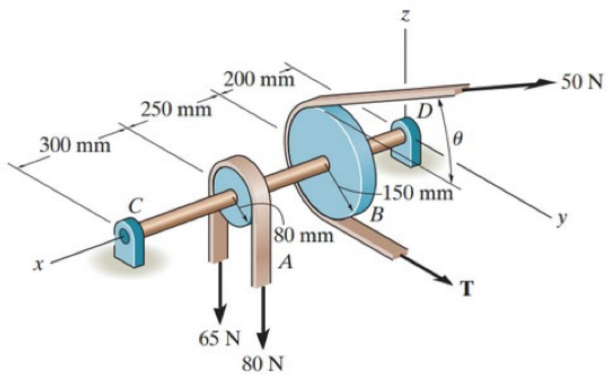
Lastly, R_A can be found to be

$$R_A = 833{,}85\,\mathrm{N} - 288{,}64\,\mathrm{N} - 332{,}17\,\mathrm{N} = 213{,}04\,\mathrm{N}.$$

Hence the three reaction forces will have sizes $\underline{\underline{R_A = 213\,\mathrm{N},\ R_B = 332\,\mathrm{N},\ \text{and}\ R_C = 289\,\mathrm{N}}}$ and they will all be acting vertically upwards in the $+z$-direction. This result makes sense as so far as the reaction force being highest for the caster closest to the crate (B), a little smaller for the sole caster a little further from the crate (C) and smallest for the caster at A, which is both somewhat far from the crate and more supported by the caster at B than was the case for the caster at C.

Exercise P6-2

Both pulleys are fixed to the shaft and as the shaft turns with constant angular velocity the power of pulley A is transmitted to pulley B. Determine the horizontal tension $\mathbf{T}$ in the belt on pulley B and the x, y, z components of reaction of the journal bearing C and thrust bearing D if $\theta = 45°$. The bearings are in proper alignment and exert only force reactions on the shaft.



As the angular velocity of the shaft is constant the system must be in dynamic equilibrium. For dynamic equilibrium to be possible the net moment must be zero about any axis must be zero.

$$\sum M = 0.$$

The moment about the x-axis at pulley A, $(M_A)_x$ is simply the sum of the two opposing moments applied here, as

$$(M_A)_x = 65\,\text{N} \cdot 80\,\text{mm} - 80\,\text{N} \cdot 80\,\text{mm} = (65\,\text{N} - 80\,\text{N}) \cdot 80\,\text{mm} = -1{,}20\,\text{N\,m}.$$

Following the same procedure, we can write an expression for the moment about the x-axis at pulley B, as

$$(M_B)_x = (T - 50\,\text{N}) \cdot 150\,\text{mm}.$$

These two moments are the only moments applied. Hence, the total moment about the x-axis is

$$\sum M_x = (M_A)_x + (M_B)_x = 0.$$

Substituting in our two expressions for $(M_A)_x$ and $(M_B)_x$ we get

$$\begin{aligned}(T - 50\,\text{N}) \cdot 150\,\text{mm} &= 1{,}20\,\text{N\,m}\\ T &= \frac{1{,}20\,\text{N\,m}}{150\,\text{mm}} + 50\,\text{N}\\ T &= 58{,}0\,\text{N}.\end{aligned}$$

This tension $T = 58{,}0\,\text{N}$ is acting vertically in the positiv y-direction and therefore the horizontal tension $\mathbf{T} = 58{,}0\,\text{N}\,\hat{j}$ in the belt on pulley B.

Now to find the bearing reactions we will utilize a force equilibrium in each coordinate direction and a moment equilibrium about the y, and z axes (no moment is produced on the shaft in the x direction by the reaction forces). The moment equilibrium will be found relative to bearing D. I.e.

$$\sum \mathbf{F} = 0$$
$$\sum (M_D)_y = 0$$
$$\sum (M_D)_z = 0.$$

Before the force equilibrium is applied, we must first determine the forces acting on the system. In this case, we have an unknown reaction force at each bearing, $\mathbf{R}_C = C_x \cdot \hat{\imath} + C_y \cdot \hat{\jmath} + C_z \cdot \hat{k}$ and $\mathbf{R}_D = D_x \cdot \hat{\imath} + D_y \cdot \hat{\jmath} + D_z \cdot \hat{k}$, an applied force on pulley A in the negative z-direction, $\mathbf{F}_A = (F_A)_z \cdot \hat{k}$, and an applied force on pulley B in the positive y and z-directions $\mathbf{F}_B = (F_B)_y \cdot \hat{\jmath} + (F_B)_z \cdot \hat{k}$.

Using these expressions for the applied forces and bearing reactions we can write out the force equilibrium in Cartesian coordinates as

$$\begin{aligned}
F_x &= C_x + D_x = 0 \\
F_y &= C_y + D_y + (F_B)_y = 0 \\
F_z &= C_z + D_z + (F_A)_z + (F_B)_z = 0.
\end{aligned}$$

As no external forces are applied in the x-direction no reaction force can arise, hence $C_x = D_x = 0$. Using trigonometry we can substitute the unknown force components in the expressions for F_y and F_z as

$$C_y + D_y + (T + \cos\theta \cdot 50\,\mathrm{N}) = 0$$
$$C_z + D_z + ((-65\,\mathrm{N}) + (-80\,\mathrm{N})) + (\sin\theta \cdot 50\,\mathrm{N}) = 0.$$

Substituting in the known value of T and θ in these expressions we get

$$\begin{aligned}
C_y + D_y + (58\,\mathrm{N} + \cos 45^\circ \cdot 50\,\mathrm{N}) &= 0 \\
\implies C_y + D_y &= -93{,}355\,\mathrm{N} \\
C_z + D_z + (-145\,\mathrm{N}) + \sin 45^\circ \cdot 50\,\mathrm{N} &= 0 \\
\implies C_z + D_z &= 109{,}644\,\mathrm{N}.
\end{aligned}$$

Now to find C_y and C_z we will utilize moment equilibrium about bearing D. In general for a force that acts at some distance x and has only y or z components the momentum is given by

$$\mathbf{M} = \mathbf{r} \times \mathbf{F} = (x, 0, 0) \times (0, F_y, F_z) = (0, -xF_z, xF_y)\,.$$

Hence,

$$\begin{aligned}
\sum (M_D)_z &= 0 \\
\implies x_C C_y - x_B (F_B)_y &= 0 \\
\implies C_y &= \frac{x_B}{x_C} (F_B)_y \\
&= \frac{0{,}2\,\mathrm{m}}{0{,}75\,\mathrm{m}} \cdot (-93{,}355\,\mathrm{N}) = -24{,}889\,\mathrm{N}
\end{aligned}$$

and

$$\begin{aligned}
\sum (M_D)_y &= 0 \\
\implies -x_C C_z - x_A (-145\,\mathrm{N}) - x_B (50\,\mathrm{N} \sin\theta) &= 0 \\
\implies C_z &= \frac{x_A 145\,\mathrm{N} - x_B (50\,\mathrm{N} \sin\theta)}{x_C} \\
&= \frac{0{,}45\,\mathrm{m} \cdot 145\,\mathrm{N} - 0{,}200\,\mathrm{m}\,(50\,\mathrm{N} \cdot \sin 45^\circ)}{0{,}75\,\mathrm{m}} = 77{,}57\,\mathrm{N}.
\end{aligned}$$

Finally, the reactions in bearing D can be found using the force equilibrium from above as

$$\begin{aligned} C_z + D_z &= 109{,}644\,\mathrm{N} \\ \implies D_z &= 109{,}644\,\mathrm{N} - C_z = 109{,}644\,\mathrm{N} - 77{,}57\,\mathrm{N} \\ &= 32{,}074\,\mathrm{N} \end{aligned}$$

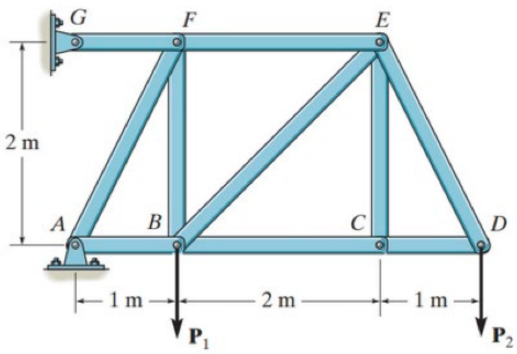
and

$$\begin{aligned} C_y + D_y &= -93{,}355\,\mathrm{N} \\ \implies D_y &= -93{,}355\,\mathrm{N} - C_y = -93{,}355\,\mathrm{N} - (-24{,}889\,\mathrm{N}) \\ &= -68{,}466\,\mathrm{N}. \end{aligned}$$

Therefore, the reaction at the bearings are $\underline{\underline{\mathbf{R}_C = (0; -24{,}889\,\mathrm{N}; 77{,}57\,\mathrm{N})}}$ and $\underline{\underline{\mathbf{R}_D = (0; -68{,}466\,\mathrm{N}; 32{,}074\,\mathrm{N})}}$ and the horizontal tension $\underline{\underline{\mathbf{T} = 58{,}0\,\mathrm{N}\,\hat{j}}}$. The fact that the reaction at the bearings is in the $-y$ and $+z$ directions makes sense considering that the applied loads are mainly in the $+y$ and $-z$ directions.

Exercise P7-1

Determine the force in each member of the truss and state if the members are in tension or compression. Set $P_1 = 8\,\text{kN}$, $P_2 = 12\,\text{kN}$.



To be able to use the method of joints, we must start at a joint with at least one known force and at most two unknown forces. For this joint we choose joint D for which a free-body diagram has been constructed on Figure 0.3. Based on this we can use the equilibrium equations to get

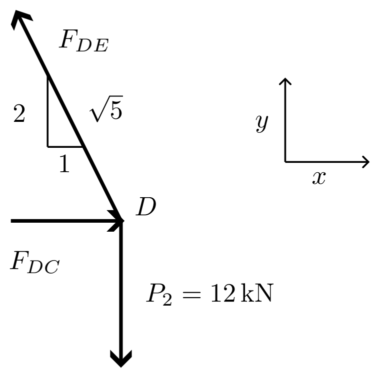


Figure 0.3: Free-body diagram of joint D.

$$\begin{aligned} F_y &= 0 \\ -12\,\text{kN} + \frac{2}{\sqrt{5}} F_{DE} &= 0 \\ \frac{2}{\sqrt{5}} F_{DE} &= 12\,\text{kN} \\ F_{DE} &= \frac{12\,\text{kN}}{2} \cdot \sqrt{5} = 13{,}42\,\text{kN} \end{aligned} \qquad \begin{aligned} F_x &= 0 \\ F_{DC} - \frac{1}{\sqrt{5}} F_{DE} &= 0 \\ F_{DC} &= \frac{1}{\sqrt{5}} F_{DE} \\ F_{DC} &= \frac{12\,\text{kN}}{2} = 6\,\text{kN}. \end{aligned}$$

Here DE is in tension and DC is in compression. Now we can continue to joint C shown on Figure 0.4.

We quickly see that $F_{CE} = 0$ and that CE therefore is a zero-force member as this is the only force acting in the y-direction. For the force equilibrium in the x-direction we have

$$\sum F_x = 0 \implies F_{CB} + F_{DC} = 0 \implies F_{CB} = -F_{DC} = -(-6\,\text{kN}) = 6\,\text{kN}.$$

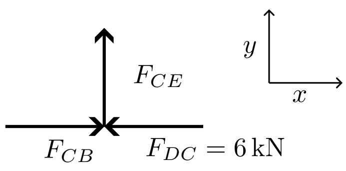


Figure 0.4: Free-body diagram of joint C.

Where CB is in compression.

Now we will move on to joint E shown on Figure 0.5.

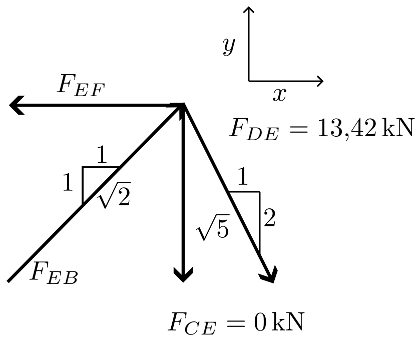


Figure 0.5: Free-body diagram of joint E.

Here we can once again employ the force balance. We start in the y-direction as

$$\sum F_y = 0$$
$$\frac{1}{\sqrt{2}}F_{EB} - \frac{2}{\sqrt{5}} \cdot 13{,}42\,\mathrm{kN} = 0$$
$$F_{EB} = \frac{2\sqrt{2}}{\sqrt{5}} \cdot 13{,}42\,\mathrm{kN} = 16{,}98\,\mathrm{kN}.$$

And in the x-direction we get

$$\sum F_x = 0$$
$$F_{EF} + \frac{1}{\sqrt{2}} \cdot 16{,}98\,\mathrm{kN} + \frac{1}{\sqrt{5}} \cdot 13{,}42\,\mathrm{kN} = 0$$
$$F_{EF} = -\left(\frac{1}{\sqrt{2}} \cdot 16{,}98\,\mathrm{kN} + \frac{1}{\sqrt{5}} \cdot 13{,}42\,\mathrm{kN}\right) = -18{,}00\,\mathrm{kN}.$$

Here, EF is in tension and EB is in compression.

Now we can move on to joint B shown on Figure 0.6.

We start by setting up the force balance in the x-direction as

$$\sum F_x = 0$$
$$F_{BA} - 6\,\mathrm{kN} - \frac{1}{\sqrt{2}} \cdot 16{,}98\,\mathrm{kN} = 0$$
$$F_{BA} = 6\,\mathrm{kN} + \frac{1}{\sqrt{2}} \cdot 16{,}98\,\mathrm{kN} = 18{,}0\,\mathrm{kN}.$$

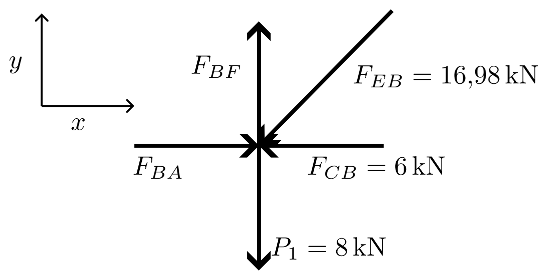


Figure 0.6: Free-body diagram of joint B.

Where BA is in compression.

Now we consider the y-direction and get

$$\sum F_y = 0$$
$$F_{BF} - \frac{1}{\sqrt{2}} \cdot 16{,}98\,\mathrm{kN} - 8\,\mathrm{kN} = 0$$
$$F_{BF} = 8\,\mathrm{kN} + \frac{1}{\sqrt{2}} \cdot 16{,}98\,\mathrm{kN} = 20\,\mathrm{kN}.$$

Where BF is in tension.

Now we can move onto joint F shown on Figure 0.7

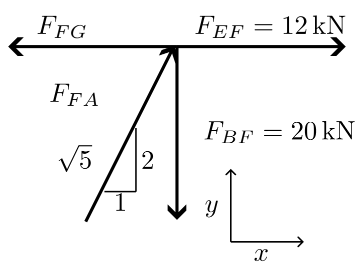


Figure 0.7: Free-body diagram of joint F.

We start in the y-direction and get

$$\sum F_y = 0$$
$$\frac{2}{\sqrt{5}} F_{FA} - 20\,\mathrm{kN} = 0$$
$$F_{FA} = \frac{\sqrt{5} \cdot 20\,\mathrm{kN}}{2} = 22{,}36\,\mathrm{kN}.$$

Here, FA will be in compression. Moving on to the x-direction we get

$$\sum F_x = 0$$
$$F_{FG} + 12\,\mathrm{kN} + \frac{1}{\sqrt{5}} \cdot 22{,}36\,\mathrm{kN} = 0$$
$$F_{FG} = -12\,\mathrm{kN} - \frac{1}{\sqrt{5}} \cdot 22{,}36\,\mathrm{kN} = -22\,\mathrm{kN}.$$

Here FG will be in tension. All in all, the following members are in tension:

$$F_{DE} = 13{,}4\,\mathrm{kN}$$
$$F_{EF} = 18{,}0\,\mathrm{kN}$$
$$F_{BF} = 20{,}0\,\mathrm{kN}$$
$$F_{FG} = 22{,}0\,\mathrm{kN}$$

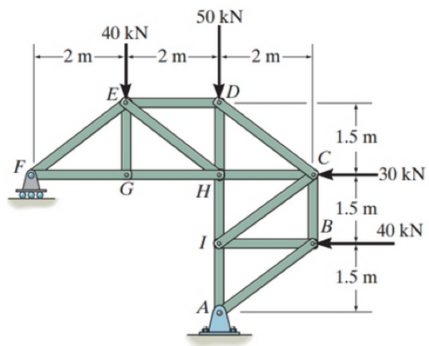
these are in compression:

$$F_{DC} = 6{,}00\,\mathrm{kN}$$
$$F_{CB} = 6{,}00\,\mathrm{kN}$$
$$F_{EB} = 17{,}0\,\mathrm{kN}$$
$$F_{BA} = 18{,}0\,\mathrm{kN}$$
$$F_{FA} = 22{,}4\,\mathrm{kN}$$

and $F_{CE} = 0$ is a zero-force member.

Exercise P8-1

Determine the force in members ED, EH, and GH of the truss and state if the members are in tension or compression.



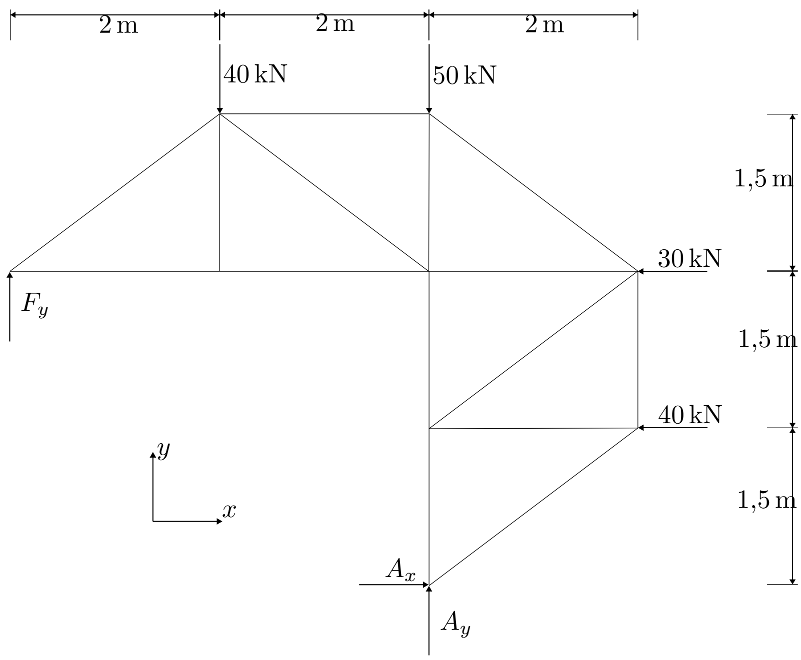


Figure 0.8: Free-body diagram for the entire truss.

First we will apply the equilibrium equations on the entire structure shown on Figure 0.8 to determine the reactions at F and A. Here we will use the forces in the x and y- directions as well as the moment about A. I.e.

$$\sum F_x = 0$$
$$\sum F_y = 0$$
$$\sum M_A = 0.$$

Only A can supply a reaction force in the x-direction as F is a roller. Therefore the sum of forces in the x-direction can be used to find this reaction as:

$$\sum F_x = A_x - 30\,\mathrm{kN} - 40\,\mathrm{kN} = 0 \implies A_x = 70\,\mathrm{kN}.$$

The sum of moments about point A can be written as

$$\sum M_A = 1{,}5\,\mathrm{m} \cdot 40\,\mathrm{kN} + 3{,}0\,\mathrm{m} \cdot 30\,\mathrm{kN} + 2{,}0 \cdot 40\,\mathrm{kN} - 4{,}0\,\mathrm{m} \cdot F_y = 0.$$

Solving this for F_y we get

$$F_y = \frac{1{,}5\,\mathrm{m} \cdot 40\,\mathrm{kN} + 3{,}0\,\mathrm{m} \cdot 30\,\mathrm{kN} + 2{,}0\,\mathrm{m} \cdot 40\,\mathrm{kN}}{4{,}0\,\mathrm{m}} = 57{,}5\,\mathrm{kN}.$$

Now we only need to find the reaction in the y-direction for A. This can be done with the force equilibrium in the y-direction as

$$\sum F_y = 57{,}5\,\mathrm{kN} - 40\,\mathrm{kN} - 50\,\mathrm{kN} + A_y = 0 \implies A_y = 32{,}5\,\mathrm{kN}.$$

Now that the reaction forces are found we can use the method of sections as shown on Figure 0.9.

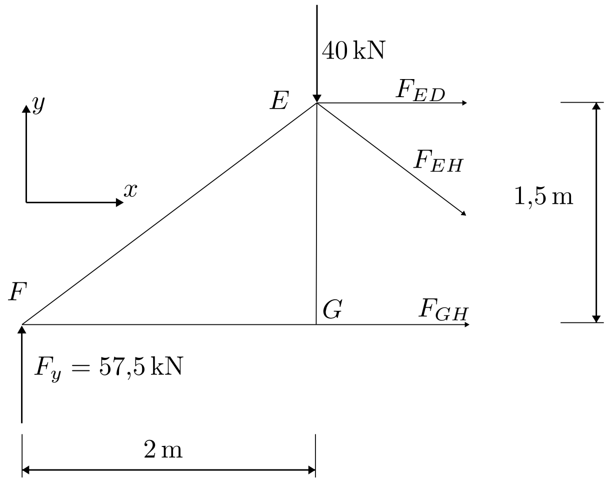


Figure 0.9: Free-body diagram of a section of the truss.

We start by considering the moment about joint E. The external force and forces F_{ED} and F_{EH} all pass through E, therefore the only forces capable of inducing a moment about E is F_y and F_{GH}. For moment equilibrium to hold about E we therefore have

$$\sum M_E = 0$$

$$F_{GH} \cdot 1{,}5\,\mathrm{m} - 57{,}5\,\mathrm{kN} \cdot 2{,}0\,\mathrm{m} = 0$$

$$F_{GH} = \frac{57{,}5\,\mathrm{kN} \cdot 2{,}0\,\mathrm{m}}{1{,}5\,\mathrm{m}} = 76{,}7\,\mathrm{kN}.$$

As this is positive the direction of F_{GH} (tension in GH) assumed on the free body diagram is true.

Now all forces acting in the y-direction are known. Hence, a force equilibrium can easily be made in this direction (as we realize EH constitutes the hypotenuse of a 3-4-5 triangle).

$$\begin{aligned}\sum F_y &= 0\\ 57{,}5\,\mathrm{kN} - 40\,\mathrm{kN} - \frac{3}{5}F_{EH} &= 0\\ F_{EH} &= \frac{5}{3}\,(17{,}5\,\mathrm{kN}) = 29{,}2\,\mathrm{kN}.\end{aligned}$$

As this is also positive our assumption of EH being in tension holds.

Now, we only need to find F_{ED}. For this we will use the force equilibrium in the x-direction as

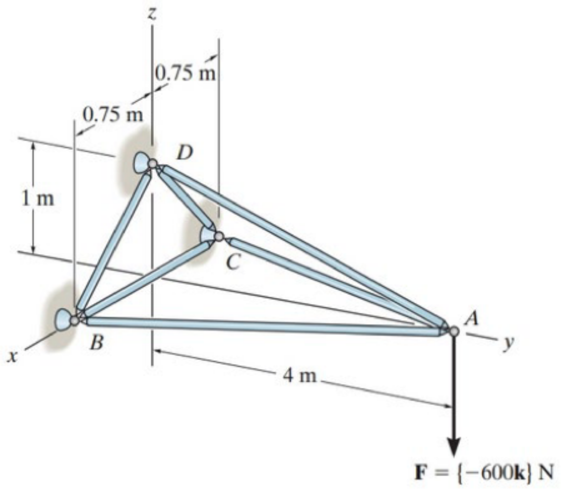
$$\begin{aligned}\sum F_x &= 0\\ F_{ED} + F_{GH} + \frac{4}{5}F_{EH} &= 0\\ F_{ED} &= -\left(F_{GH} + \frac{4}{5}F_{EH}\right) = -100\,\mathrm{kN}.\end{aligned}$$

As this is negative the assumption of ED being in tension, as shown on Figure 0.9 is *not* true, hence ED must be in compression.

All in all, $\underline{\underline{F_{GH} = 76{,}7\,\mathrm{kN} \text{ and } F_{EH} = 29{,}2\,\mathrm{kN} \text{ are in tension and } F_{ED} = 100\,\mathrm{kN} \text{ is in compression.}}}$

Exercise P8-2

Determine the force in members AB, AD, and AC of the space truss and state if the members are in tension or compression.



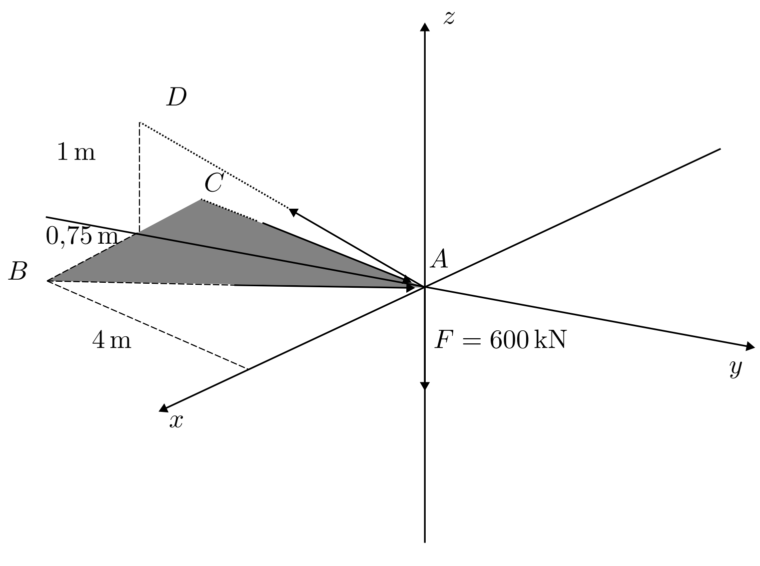


Figure 0.10: Free-body diagram of the space truss

We use the method of joints starting at point A as shown on Figure 0.10. At this point, only the applied force $\mathbf{F}$ and the z-component of F_{AD} acts in the z-direction. To find the z-component of F_{AD} we first must find the length $|AD|$. This can be found using the Pythagorean theorem as

$$|AD| = \sqrt{(4\,\mathrm{m})^2 + (1\,\mathrm{m})^2} = \sqrt{17}\,\mathrm{m}.$$

Now an expression for the z-component of F_{AD} can easily be found by trigonometry as

$$(F_{AD})_z = \frac{1}{\sqrt{17}} \cdot F_{AD}.$$

Now we can write out the force equilibrium in the z-direction as

$$\begin{aligned}\sum F_z &= 0\\ -600\,\mathrm{N} + \frac{1}{\sqrt{17}} \cdot F_{AD} &= 0\\ F_{AD} &= 600\,\mathrm{N} \cdot \sqrt{17} = 2473{,}86\,\mathrm{N}.\end{aligned}$$

To balance the applied force out AD must be in tension.

Now we only need to find the forces F_{AC} and F_{AB} (both of which must be in compression to balance the applied force and F_{AD}). To do this we will use the force equilibria in the x and y-directions. Before we can do this, however, we must first determine the length $|AC| = |AB|$ using the Pythagorean theorem.

$$|AC| = |AB| = \sqrt{(0{,}75\,\mathrm{m})^2 + (4\,\mathrm{m})^2} = 4{,}0697\,\mathrm{m}.$$

Using this length we can write expressions for the x and y components of F_{AB} and F_{AC}. We will also find the y component of F_{AD}

$$\begin{aligned}(F_{AB})_x &= \frac{0{,}75}{4{,}0697} \cdot F_{AB}\\ (F_{AC})_x &= \frac{0{,}75}{4{,}0697} \cdot F_{AC}\\ (F_{AB})_y &= \frac{4}{4{,}0697} \cdot F_{AB}\\ (F_{AC})_y &= \frac{4}{4{,}0697} \cdot F_{AC}\\ (F_{AD})_y &= \frac{4}{\sqrt{17}} F_{AD}.\end{aligned}$$

Now a force equilibrium can be made in the x and y directions as

$$\begin{aligned}\sum F_x &= 0\\ \frac{0{,}75}{4{,}0697} \cdot (F_{AB} - F_{AC}) &= 0\\ F_{AB} &= F_{AC}\\ \sum F_y &= 0\\ -\frac{4}{4{,}0697}\left(F_{AB} + F_{AC}\right) - \frac{4}{\sqrt{17}} \cdot F_{AD} &= 0\\ -\frac{4}{4{,}0697}\left(2F_{AB}\right) &= \frac{4 \cdot F_{AD}}{\sqrt{17}}\\ F_{AB} &= -\frac{4{,}0697 \cdot 2473{,}86\,\mathrm{N}}{2 \cdot \sqrt{17}} = -1221\,\mathrm{N}\\ F_{AC} &= F_{AB} = -1221\,\mathrm{N}.\end{aligned}$$

Therefore $\underline{\underline{F_{AD} = 2{,}47\,\mathrm{kN} \text{ is in tension and } F_{AB} = F_{AC} = 1{,}22\,\mathrm{kN} \text{ is in compression}}}$.